

✓ 10-DAY SYSTEM TESTING PROTOCOL

For the Web-Based Ethiopian Construction Data Management System (ECDMS)
Based on the SRS (Nov 04, 2024 Draft)
For: ERA, ECA, MoUI Volunteer Testing Teams

1. Testing Framework & Basis

The testing protocol is derived from the SRS’s defined components:

Relevant SRS Sections Used

- System Features / Functional Requirements
- System Architecture & Technology Stack
- Security Requirements
- User Roles & Access Levels
- Use Case Flows
- Integration Requirements (SMS, Email, APIs)
- Data Models (CRUD, Validation, Workflows)

2. Test Crew Structure (As Provided)

The protocol assumes the following stakeholder testers:

Stakeholder	Roles in Testing
Ethiopian Roads Administration (ERA)	Road project workflows, supervision, materials, reports
Ethiopian Construction Authority (ECA)	Building works, compliance, document control
MoUI Central PMO	Governance, master data, final UAT validation
Contractors	Material, manpower, equipment reporting
Consultants	Design approvals, inspections, RFIs

3. Testing Approach Overview

The 10-day approach follows **compressed national testing methodology**, as required:

Environment Setup → Functional Testing → Master Data Testing → Workflow Testing
→ Integration Testing → Performance → Security → UAT → Sign-off

4. Detailed 10-Day Testing Schedule

DAY 1 – TEST ENVIRONMENT SETUP & ORIENTATION

Tasks:

- Deploy latest ECDMS testing build
- Create test accounts per stakeholder role
- Import sample master datasets:
 - Manpower
 - Material
 - Stakeholder List
 - Projects
 - Documents
 - Equipment Resources
- Conduct a 2-hour orientation (MoUI → all teams)
- Distribute test cases & defect reporting templates
- Smoke test from MoUI ICT

Outputs:

- Test environment ready
- Accounts active
- Kick-off report

DAY 2 – USER ACCESS & SECURITY ROLE TESTING

Aligns with SRS's RBAC requirements.

Tests:

- Login, logout, password reset
- Role-based access (Admin vs Consultant vs Contractor vs ERA/ECA Officer)
- Unauthorized access attempts
- Menu visibility validation
- Access to restricted documents & modules
- Session timeout & re-authentication

Expected Outcome:

System respects SRS-defined user roles and access hierarchy.

DAY 3 — FUNCTIONAL TESTING (MODULE 1-3)

Based on the SRS functional decomposition.

1. Manpower Management

- Add/Edit/Delete manpower
- Professional category validation
- Linking manpower to contractors & projects

2. Material Management

- Add materials, edit units, classification checks
- Approved supplier linkage
- Importing bulk materials

3. Resource/Equipment Registry

- Register equipment type, capacity, ownership
- Equipment assignment workflows

Results:

All master data modules operate as defined in SRS.

DAY 4 — FUNCTIONAL TESTING (MODULE 4-6)

4. Stakeholder Registry

- Entity types (Consultant, Contractor, ERA/ECA, MoUI)
- License validation
- Status changes (Active, Expired, Suspended)

5. Project Registry

- Create/edit projects
- Assign consultant/contractor
- Baseline data validation

6. Document Management

- Upload, versioning, preview
- Category validation (Technical, Contractual, Financial)

- Approval workflows
- Document access permissions

DAY 5 — INTEGRATED WORKFLOW TESTING

Based on SRS use-case flow diagrams.

Workflows to Test:

1. Daily Construction Report → Supervisor Review → Approval
2. RFI Creation → Consultant Response → Closure
3. Material Delivery → Quality Check → Stock Update
4. Document Submission → Comment → Version Update → Final Approval

Validation:

- Status tracking
- Notifications (email/SMS)
- Linked datasets update (materials, manpower, equipment)

DAY 6 — ADVANCED WORKFLOW TESTING (ERA + ECA)

ERA-specific tests:

- Road project dashboards
- Progress reporting
- Material usage logs (asphalt, aggregates)
- Site supervision templates

ECA-specific tests:

- Building compliance workflows
- Design review & approval
- Contractor performance evaluation

DAY 7 — INTEGRATION TESTING

According to SRS integration specifications.

Areas:

- SMS gateway
- Email notifications
- API integrations (if enabled: licensing, project import, GIS)
- Document storage integration (cloud/local server)

Tests:

- Trigger notifications from various modules
- Ensure correct API responses
- Error handling when external service is down

DAY 8 — PERFORMANCE & LOAD TESTING

As required by SRS non-functional constraints.

Simulations:

- 100–200 concurrent users
- Bulk document uploads (50MB+)
- Rapid data entry bursts
- Load on dashboards & queries

Metrics:

- Response time
- Page load speed
- Upload success/failure
- Timeout behavior

DAY 9 — SECURITY & VULNERABILITY TESTING

Aligned with SRS security requirements.

Tests:

- SQL injection
- XSS injection
- Broken access control attempts
- URL manipulation
- File upload sanitization
- Data encryption verification
- Audit trail validation

DAY 10 — USER ACCEPTANCE TESTING (UAT) & SIGN-OFF

Participants:

- ERA
- ECA
- MoUI
- Contractors
- Consultants

Activities:

- Test real-world construction scenarios
- Validate data reports
- Confirm compliance with SRS needs
- Approve or reject modules
- Document remaining issues for future patches

Deliverables:

- UAT Report
- Final Bug Matrix
- Go-Live Recommendation

5. Defect Reporting Protocol

Each tester logs findings in a national defect log with:

- Test Case ID
- Module
- Severity: Critical / High / Medium / Low
- Steps to reproduce
- Expected vs actual results
- Screenshot/video
- Stakeholder name

MoUI QA team consolidates daily.

6. Testing Completion Criteria

ECDMS testing is considered successful when:

- 100% of test cases executed
- All **critical** issues fixed
- All **high** issues have fixes or workarounds
- 80–90% features accepted in UAT
- All workflows validated
- Integration services stable

7. Optional Add-ons (I can prepare on request)

ECDMS Testing Optional Add-Ons Package

This document contains all templates required under Item No. 7: 1. Test Case Matrix (TCM) 2. Defect Log Template 3. Daily Testing Schedule (10-Day Gantt Format) 4. Stakeholder Testing Responsibility Matrix 5. 10-Day Testing Plan (Formatted) 6. Workflow Diagrams (ASCII-based for clarity)

1. TEST CASE MATRIX (TCM TEMPLATE)

Test Case ID	Module	Test Scenario	Pre-Conditions	Steps	Expected Result	Actual Result	Status	Tester	Severity
TC-UM-001	User Management	Login with valid credentials	User account active	1. Open Login Page2. Enter credential3. Click Login	User successfully logs in				Medium
TC-MM-002	Material Registry	Add new material	User logged in as Admin	1. Open Material	Material saved and appears in list				High

Test Case ID	Module	Test Scenario	Pre-Conditions	Steps	Expected Result	Actual Result	Status	Tester	Severity
TC-DM-015	Document Mgmt	Upload new document	Project exists	al Reg istr y2. Cli ck Ad d3. Fill for m4. Sav e					
				1. Op en Do cu me nts	Document stored and version = 1				Critical
				2. Up loa d file 3. Sub mit					

(Use this template to expand to all modules.)

2. DEFECT LOG TEMPLATE

Defect ID	Module	Description	Steps to Reproduce	Severity	Priority	Reported By	Date	Assigned To	Status	Evidence
DF-01	Document Module	Document failing to upload	1. Open Docs2. Upload 25MB file	Critical	P1	ERA Tester	2025-11-18	Dev Team	Open	Screenshot

Day 1 – Environment setup, account creation, smoke test

Day 2 – User access, permissions, role validation

Day 3 – Manpower, Material, Resource registries

Day 4 – Stakeholder, Project, Document registries

Day 5 – Workflow testing (Daily Reports, RFIs)

Day 6 – ERA/ECA specific workflows

Day 7 – Integration testing (SMS, Email, APIs)

Day 8 – Performance & Load tests

Day 9 – Security, vulnerability scans, access tests

Day 10 – UAT, issue closure, final evaluation and sign-off

6. WORKFLOW DIAGRAMS (ASCII FORMAT)

6.1 Daily Report Workflow

Contractor → Submit Daily Report



Consultant → Review & Comment



ERA/ECA Supervisor → Final Approval



Status = Approved & Archived

6.2 Document Approval Workflow

User Uploads Document



System Assigns Version



Consultant Reviews



ERA/ECA Approves or Rejects



If Rejected → Revision Required

If Approved → Locked + Published

6.3 RFI Workflow

Contractor Creates RFI



Consultant Responds



ERA/ECA Confirms Resolution



RFI Status = Closed